US Accidents Severity
Data Preparation, Exploration and Feature Engineering

Data Science Midterm Project · Written Report
CRISP-DM phases 1-6: business understanding to feature engineering and next steps

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Dataset: US Accidents (2016-2023), Moosavi et al. [1-3], 500,000-accident sample

Code: notebooks 01-04, `src/features.py` and `run_all.py` in the project repository

Summary

Traffic centres need to know quickly which accidents will cause long delays. This project prepares a 500,000-accident sample of the US Accidents dataset to predict **Severity** (1 = small delay, 4 = long delay) using only information known when an accident is reported. We removed 11 columns that would reveal the answer, cleaned the data (495,532 accidents remain), explored it with 14 charts and 6 hypothesis tests, and built 36 model inputs. The main findings: severity is very unbalanced (80% are level 2); the **data provider** has the strongest link with severity (Cramér's V = 0.29), which means part of the label depends on who recorded it; **highway** accidents are 2.5 times more often high-impact; night and bad weather matter only a little. No model is trained yet; Section 6 proposes the modelling plan.

1. Business Understanding

1.1 Context

The US has millions of road accidents every year. Some only slow traffic for a few minutes; others close a highway for hours. Traffic management centres and navigation apps must decide, right after an accident is reported, how many responders to send and whether to reroute drivers. A tool that predicts the impact of an accident from the first report would help them act faster.

1.2 Problem statement

Given when, where and in what conditions an accident happens, can we predict how much it will slow down traffic (Severity 1-4), using only information available at the moment it is reported?

Important: in this dataset, Severity measures the **impact on traffic** (delay), not injuries. This is a **multi-class classification** problem with four very unbalanced classes.

1.3 Research questions

Table 1. Research questions. Each question is answered with charts and, where possible, a hypothesis test.

#	Research question	Answered in
RQ1	When do high-impact accidents happen (hour, weekday, day vs night)?	4.3, 4.5
RQ2	Does weather change the severity of accidents?	4.3, 4.5
RQ3	Does the road context (highway, traffic signals, junctions) change severity?	4.3, 4.5, 5.1
RQ4	Is the severity label consistent across data providers and years?	4.4, 4.5
RQ5	Which information known at report time is most related to severity, and should be given to a model?	5.3

1.4 Success criteria

For this midterm (data preparation):

- **No leakage:** every kept column must be known at report time; every removal has a written reason.
- **Traceable cleaning:** rows and columns are logged before and after each step.
- **Evidence for each research question:** a labelled chart and, where possible, a hypothesis test with an effect size (not only a p-value).
- **Justified features:** the final feature set is chosen with clear rules on training data only.
- **Reproducible:** one command (`python run_all.py`) rebuilds every result with a fixed random seed (42).

For the future model (next steps): always predicting "Severity 2" already gives about 80% accuracy, so accuracy is misleading. The model will be judged by **macro F1** (the average F1 of the 4 classes), which must be clearly above the 0.22 of that trivial model, and by the **recall of Severity 4** (the share of the worst accidents it detects), which must be clearly above 0.

2. Data Understanding

2.1 Source

The data comes from the **US Accidents** dataset by Sobhan Moosavi et al. (Ohio State University) [1-4], published on Kaggle under the CC BY-NC-SA 4.0 licence (academic use allowed with citation). The full dataset has about 7.7 million accidents from 49 US states, February 2016 to March 2023, collected in real time from traffic APIs (three data providers, called Source1-3) that broadcast traffic events reported by transportation departments, law enforcement, traffic cameras and road sensors [1]. The authors added weather observations from the nearest airport station and nearby road features from OpenStreetMap [2]. This project uses the 500,000-row random sample included in the repository (`US_Accidents_March23_sampled_500k.csv`), with one source file and no merging.

2.2 Size and structure

The sample has **500,000 rows** (one per accident) and **46 columns**: 20 text, 13 True / False, 12 number (decimal), 1 number (integer). Table 2 groups them; **Appendix A** defines every variable with its type, unit and share of missing values.

Table 2. Columns of the raw sample. Units are part of the column names, e.g. (F) = degrees Fahrenheit.

Group	Columns	Type and unit
Target	Severity	Integer 1-4 (traffic impact)
Identifiers	ID, Source	Text
Time	Start_Time, End_Time, Weather_Timestamp, Timezone	Date-time (local time), text
Location	Start_Lat, Start_Lng, End_Lat, End_Lng, Distance(mi)	Degrees; Distance in miles
Address	Street, City, County, State, Zipcode, Country, Airport_Code	Text
Description	Description	Free text written by the provider
Weather	Temperature(F), Wind_Chill(F), Humidity(%), Pressure(in), Visibility(mi), Wind_Speed(mph), Precipitation(in), Wind_Direction, Weather_Condition	°F, %, inches of mercury, miles, mph, inches; text
Road features	Amenity, Bump, Crossing, Give_Way, Junction, No_Exit, Railway, Roundabout, Station, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop	True / False (feature near the accident)
Light	Sunrise_Sunset, Civil_Twilight, Nautical_Twilight, Astronomical_Twilight	Day / Night

Table 3. The target variable. The sample keeps the class shares of the full dataset (difference below 0.1 percentage point).

Severity	Meaning	Accidents (sample)	Share (sample)	Share (full 7.7M)
1	Least impact	4,274	0.85%	0.87%
2	Moderate	398,142	79.63%	79.67%
3	Significant	84,520	16.90%	16.81%
4	Highest impact (long delay)	13,064	2.61%	2.65%

Key variables. The target is Severity. From the problem statement, the most important inputs are the time (Start_Time), the place (coordinates, state, city, street), the weather columns and the 13 road features.

2.3 Initial data quality issues

Table 4. Data quality issues found during data understanding.

Issue	What we found	Handled in
Missing values	End_Lat/End_Lng 44%, Precipitation 29%, Wind_Chill 26%, Wind_Speed 7.4%, other weather about 2%, light columns 0.3%, City/Zipcode under 0.03%	3.3
Leakage columns	End_Time, End_Lat/Lng, Distance(mi) and Description are only known after the accident	3.1
Impossible values	Temperature 207 °F and -77.8 °F, wind speed 822.8 mph, visibility 130 miles, pressure 0.12 inHg	3.4
Inconsistent categories	Wind_Direction has 24 spellings (e.g. "West" and "W", "Calm" and "CALM"); Weather_Condition has 108 labels	3.5
Mixed date formats	Start_Time is written with and without fractional seconds	3.2
Constant columns	Country is always "US", Turning_Loop is always False	3.1
Duplicates	None with the ID column; 2,884 identical rows once the ID is removed	3.2
Provider differences	The three providers use the severity levels very differently (Section 4.4)	4.4, 5

Limitations of the data. It is a sample (6.5% of the full dataset); it ends in March 2023 (only 3 months of 2023); coverage depends on the providers, so some states and years have more data; Severity is assigned by the providers, not measured in the same way by everyone.

3. Data Pre-processing

All steps are in notebook `02_cleaning.ipynb`; every step is logged in `reports/cleaning_log.md`. The data went from **500,000 × 46** to **495,532 × 36**.

3.1 Preventing data leakage: column selection

Rule: **a column is kept only if its value is known when the accident is first reported.** A model trained with "after-the-fact" columns would look excellent in testing but would be useless in real life. Table 5 shows that these columns contain the answer.

Table 5. Evidence of leakage. Leaking columns separate Severity 4 from the rest almost perfectly, so they must be removed.

Severity	Median duration (End - Start)	Mean affected distance	Description says "closed"
1	45 min	0.11 mi	2.2%
2	78 min	0.57 mi	4.7%
3	44 min	0.42 mi	5.4%
4	129 min	1.48 mi	88.5%

Table 6. The 11 removed columns. Kept with care: Source (tested later with and without), and City, County, Street, Zipcode (thousands of values, handled in feature engineering).

Removed column	Reason
End_Time	Leakage: known only after the road is cleared (long duration means severe)
End_Lat, End_Lng	Leakage: end of the affected road stretch, measured afterwards
Distance(mi)	Leakage: length of road affected is part of how severity is judged
Description	Leakage: text such as "road closed" or "lanes blocked"
ID	Identifier only, no information
Country, Turning_Loop	Constant: the same value in every row (checked in the data)
Weather_Timestamp, Timezone, Airport_Code	Redundant with Start_Time and the location columns

Leakage is also prevented in later transformations: anything that *learns* from the data (median filling, encodings, clusters, scaling, feature selection) is fitted on a training split only (Section 5).

3.2 Types and duplicates

- Start_Time converted to a date-time with `format="mixed"` to accept both formats.
- The 12 road-feature columns changed from True/False to 1/0 so models can use them.
- 2,884 exact duplicate rows removed (the same accident recorded twice would count double).

3.3 Missing values

We measured the share of empty cells per column, then chose a treatment per case (Table 7).

Table 7. Missing-value treatments. Percentages are shares of rows in the sample.

Column(s)	Missing	Treatment	Why
Wind_Chill(F)	26%	Column removed	Almost a copy of Temperature (correlation 0.994)
Precipitation(in)	29%	Filled with 0 + new flag precip_missing (1 = was empty)	Empty mostly means no rain reported; the flag keeps the information that it was guessed
City, Zipcode, light columns	0.3%	Rows removed (1,584 rows)	Too few rows to be worth guessing
Weather_Condition, Wind_Direction	2.2%	Filled with "Unknown"	Keeps the rows; "unknown" becomes its own category
Temperature, Humidity, Pressure, Visibility, Wind_Speed	2-7%	Left empty now; filled with the **median of the training data** when the model is fitted	Filling now would use test data (leakage); median is robust to the skewed values

3.4 Outliers and impossible values

We checked the ranges with summary statistics and box plots (Figure 1). Values that are physically impossible were set to empty (then treated like any missing value). Extreme but possible values were kept.

Table 8. Limits for impossible values.

Column	Allowed range	Values removed	Reason for the limit
Temperature	-50 to 130 °F	4	Record US high is 134 °F; below -50 °F is not plausible on these roads
Wind speed	0 to 200 mph	3	Stronger than any surface wind measured outside tornadoes
Visibility	0 to 100 miles	3	Weather stations do not report more than about 100 miles
Pressure	15 to 32 inHg	3	Highest US roads are about 17.5 inHg; world record high is 32.06 inHg

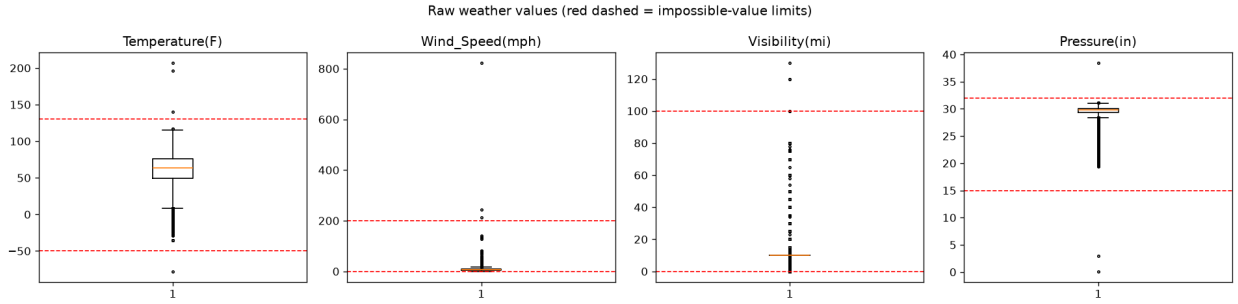


Figure 1. Weather values before cleaning. Box plots of the raw values; red dashed lines are the limits of Table 8.

3.5 Inconsistent categories

- **Wind_Direction:** 24 spellings merged into 16 compass points + CALM + VAR (+ Unknown), e.g. "West" → "W", "Calm" → "CALM", "Variable" → "VAR".
- **Weather_Condition:** 108 labels grouped into 8 groups + Unknown (new column Weather_Group): Clear, Cloudy, Rain, Heavy rain / Thunderstorm, Snow / Ice, Fog / Haze / Smoke, Windy, Other. Rules are applied in order of danger, so "Light Snow with Thunder" counts as Thunderstorm.

3.6 Effect on the dataset

Table 9. Dataset size after each step. Only 0.9% of rows were removed. Remaining empty cells: 5 weather columns (2-7%) and Street (0.1%).

Step	Rows	Columns
Raw sample	500,000	46
After removing leaking / useless columns	500,000	35
After removing duplicates	497,116	35
After missing-value treatment	495,532	35
After grouping weather (final clean data)	495,532	36

4. Exploratory Data Analysis

All charts and tests come from notebook `03_eda.ipynb` (files in `reports/figures/`). "High impact" means Severity 3 or 4. Section 4.1 describes the variables, 4.2 shows when and where accidents happen, 4.3 relates the variables to Severity (RQ1-RQ3), 4.4 checks whether the label is stable (RQ4) and 4.5 tests the main relationships statistically.

4.1 Descriptive statistics

Table 10. Summary of the numeric weather variables (clean data). Skew above 0 means a long tail of high values; below 0, a long tail of low values.

Variable	Mean	Median	Std	Min	Max	Missing	Skew
Temperature(F)	61.64	64.00	18.99	-35	117	2.0%	-0.51
Humidity(%)	64.84	67.00	22.83	1	100	2.2%	-0.40
Pressure(in)	29.54	29.86	1.01	19.36	31.15	1.7%	-3.68
Visibility(mi)	9.09	10.00	2.69	0	100	2.2%	2.32
Wind_Speed(mph)	7.68	7.00	5.27	0	142	7.4%	0.93
Precipitation(in)	0.01	0.00	0.09	0	10.13	0.0%	91.80

Table 10 shows that most accidents happen in mild, dry, clear weather (median 64 °F, 10 miles of visibility, no rain). Precipitation (skew 92) and visibility are very skewed and pressure is skewed to the left (stations at high altitude), so medians describe them better than means; this is also why missing values are filled with the median. Among the categories, Source1 reports 56% of accidents, California 23% and "Clear" weather 44%.

Figure 2 shows the target. **Interpretation:** the classes are very unbalanced: 80% Severity 2, 0.9% Severity 1 and 2.6% Severity 4. A model that always answers "2" would be 80% accurate while being useless, which is why accuracy is not used and macro F1 will be.

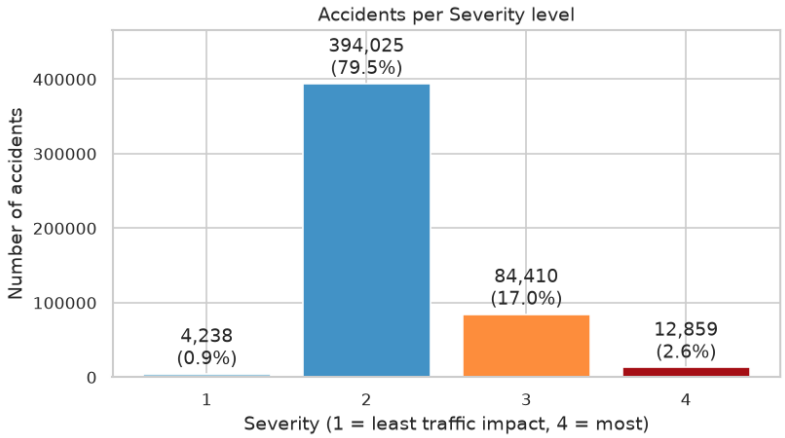


Figure 2. Accidents per Severity level. Bar heights are counts; labels give the share of all accidents.

4.2 When and where accidents happen

Before looking at severity, Figures 3-5 show how the accidents themselves are distributed in time and space.

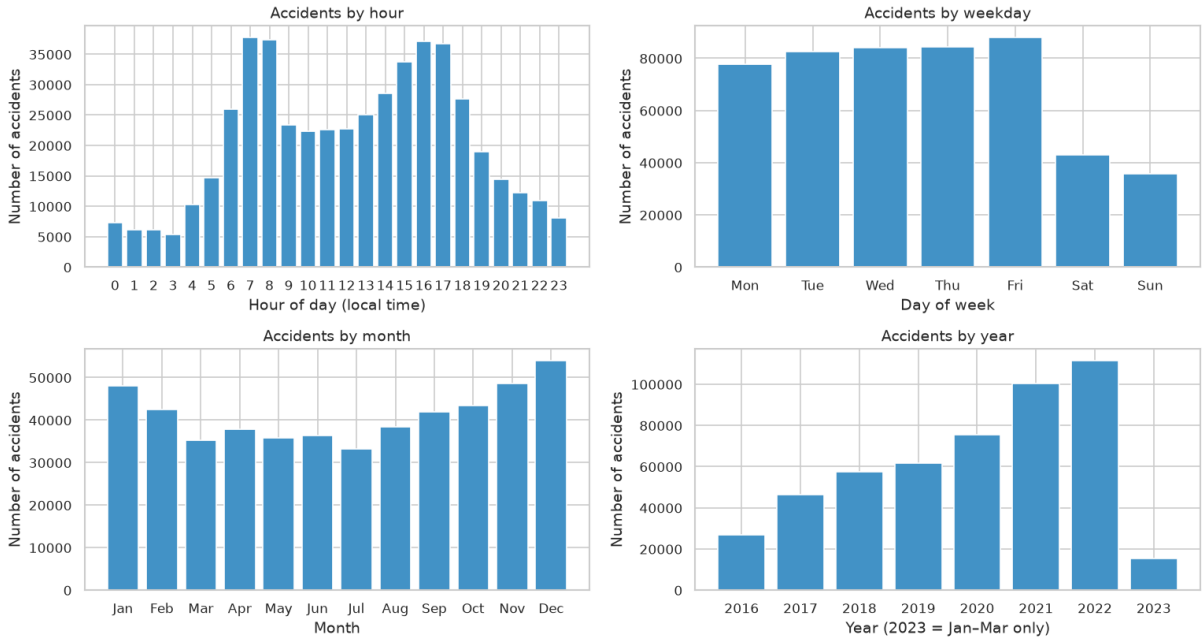


Figure 3. Number of accidents by hour, weekday, month and year. Hour is local time; 2023 contains only January to March.

Interpretation (Figure 3): accidents follow commuting: two peaks at 7-8 h and 15-17 h, and about half as many accidents on Saturdays and Sundays as on weekdays. December and January are the busiest months (darkness, winter weather, holidays). The count per year rises from 26,700 (2016) to 113,700 (2022); this mostly reflects better data collection, not more accidents, which is a first warning that the data changes over time (Section 4.4).

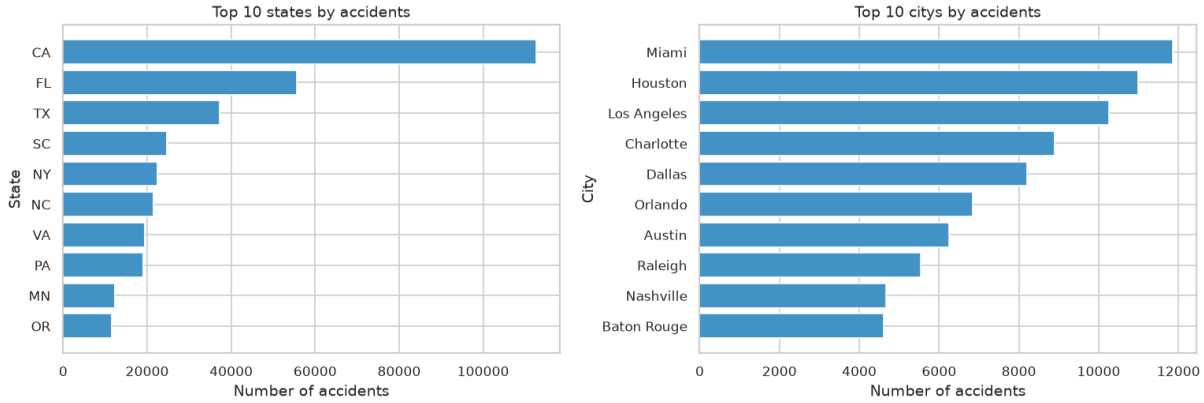


Figure 4. Top 10 states and cities by number of accidents. Horizontal bars; counts in the clean sample.

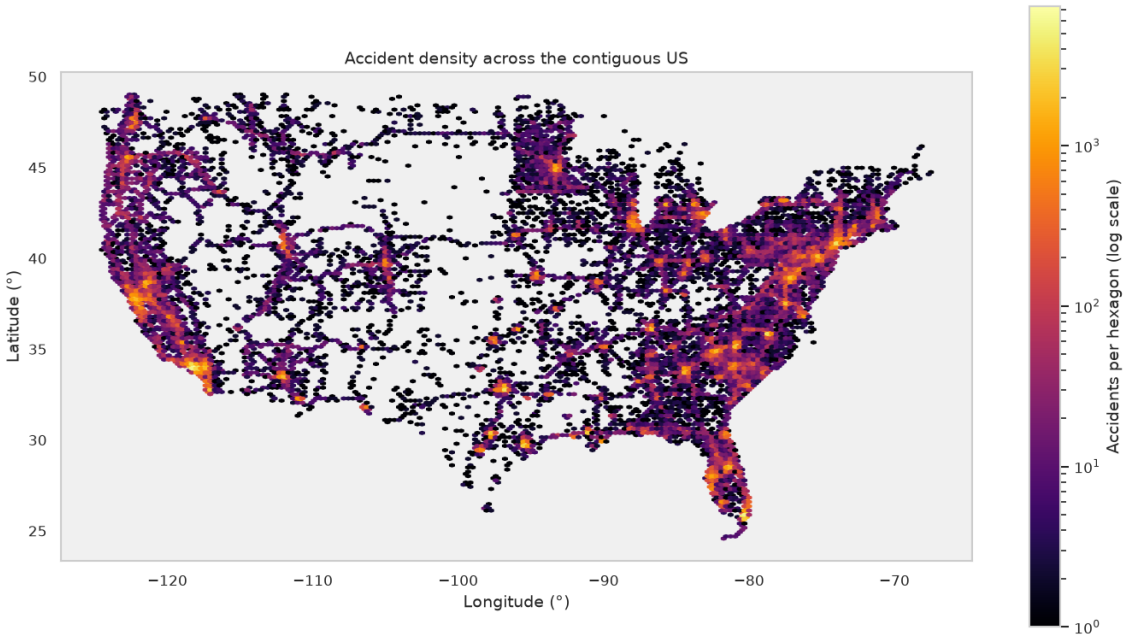


Figure 5. Accident density across the contiguous United States. Each hexagon counts the accidents inside it; the colour scale is logarithmic.

Interpretation (Figures 4-5): California alone has 23% of the accidents, then Florida (11%) and Texas (7.5%); the top cities are Miami, Houston and Los Angeles. The map follows the interstate network and large metropolitan areas and is almost empty in the Mountain West. Counts depend on population **and** on how well each provider covers an area, so location is a strong but partly biased signal.

4.3 Relationships with Severity (RQ1-RQ3)

RQ1 (time). Figure 6 shows the severity mix for each hour and Figure 7 compares day and night.

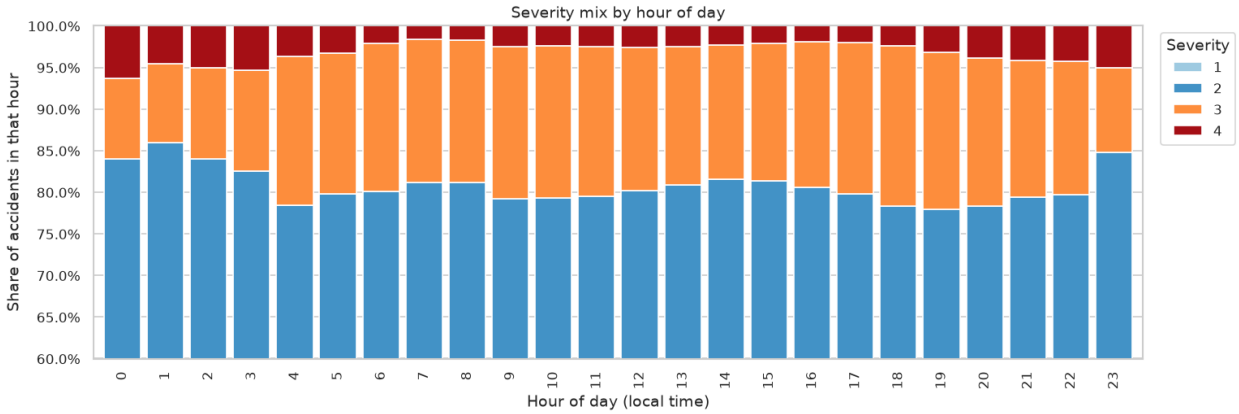


Figure 6. Severity mix by hour of the day. Each bar is one hour (local time) and sums to 100%; the y-axis starts at 60% to make the small classes visible.

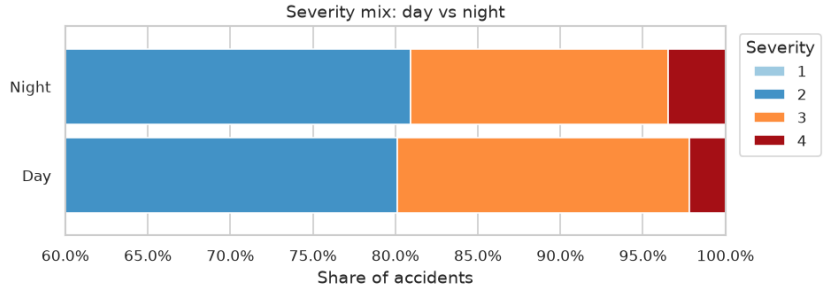


Figure 7. Severity mix by day and night. Day / night is based on sunrise and sunset; the x-axis starts at 60%.

Interpretation: rush hours have the most accidents but *not* the most severe ones: Severity 4 is lowest at 7-8 h (1.6%) and highest around midnight (5-6%). At night, Severity 4 is 1.6 times more common than by day (3.5%

vs 2.2%), while Severity 3 is slightly less common (15.6% vs 17.7%). Likely reasons: faster driving on empty highways, poorer visibility and slower response at night.

RQ2 (weather). Figure 8 compares weather groups and Figure 9 compares visibility and temperature across severity levels.

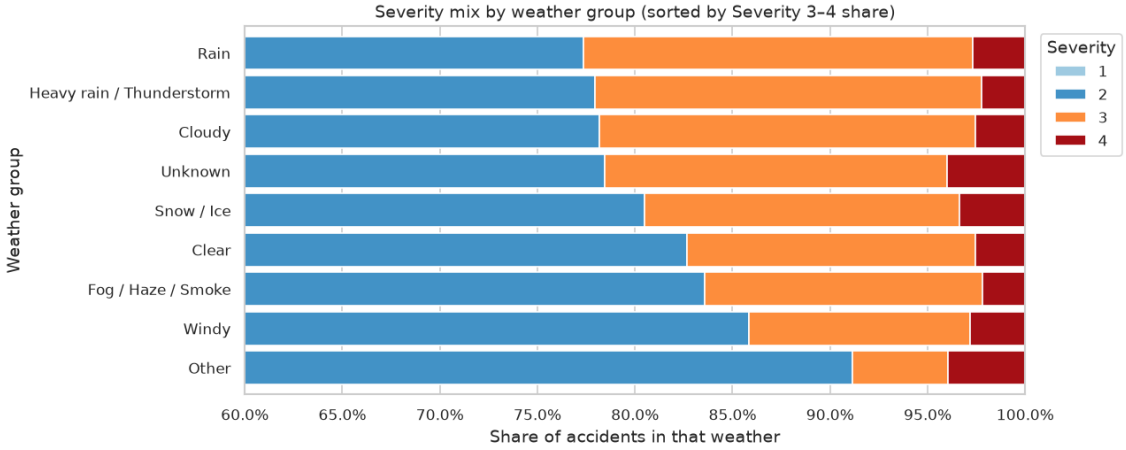


Figure 8. Severity mix by weather group. Groups are sorted by their share of Severity 3-4; the x-axis starts at 60%.

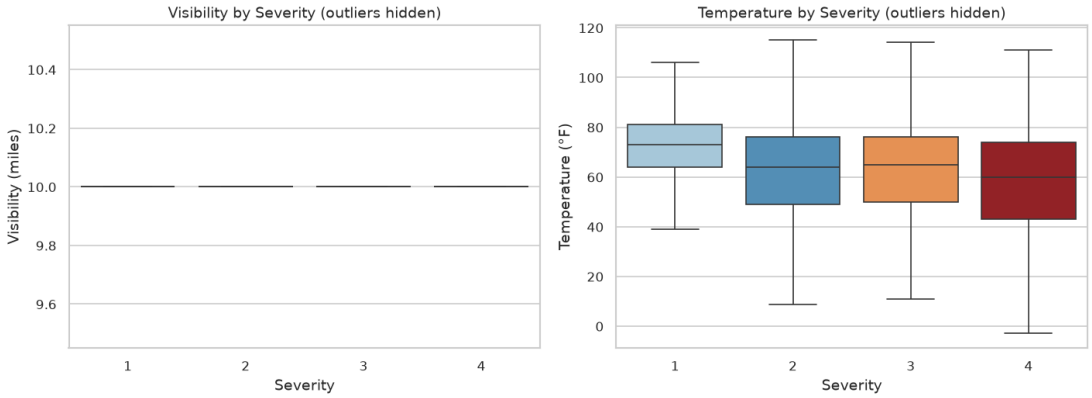


Figure 9. Visibility and temperature by Severity level. Box plots without outliers: the box covers the middle 50% of accidents and the line is the median.

Interpretation: rain and thunderstorms have about 22% high-impact accidents against 17% in clear weather, and snow/ice raises Severity 4 (3.4% vs 2.6%). Visibility is the same for every level (median 10 miles: most accidents happen in good visibility), and Severity 4 accidents happen at slightly lower temperatures (median 60 °F vs 64 °F for Severity 2). Weather matters, but much less than expected.

RQ3 (road). Figure 10 compares accidents with and without four common road features.

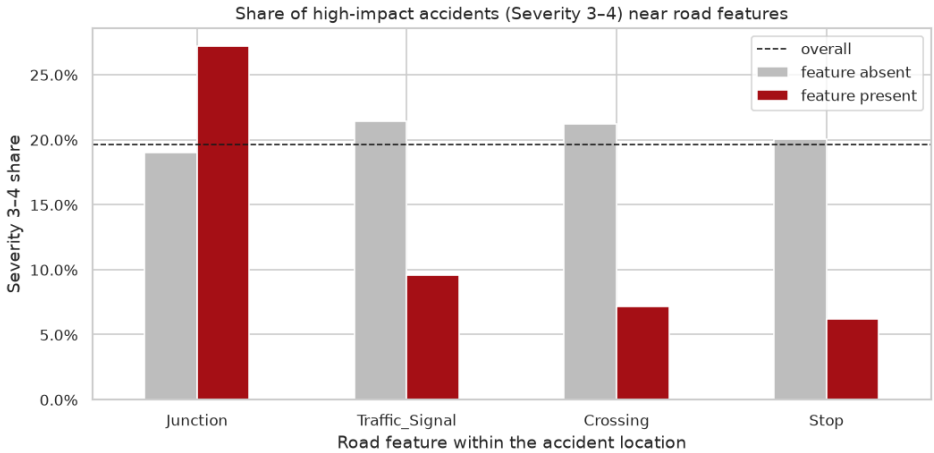


Figure 10. Share of high-impact accidents near road features. Grey: feature absent; red: feature present; dashed line: overall share (19.6%).

Interpretation: accidents near traffic signals, crossings or stop signs are 2-3 times less often high-impact (6-10% vs about 20%) because these are slow city streets. Junctions are the exception (27%) because they include highway ramps. In Section 5, highway accidents turn out to be 28% high-impact against 11% elsewhere: road type is one of the clearest signals in the data.

Correlations. Figure 11 shows the Pearson correlations between the numeric variables.

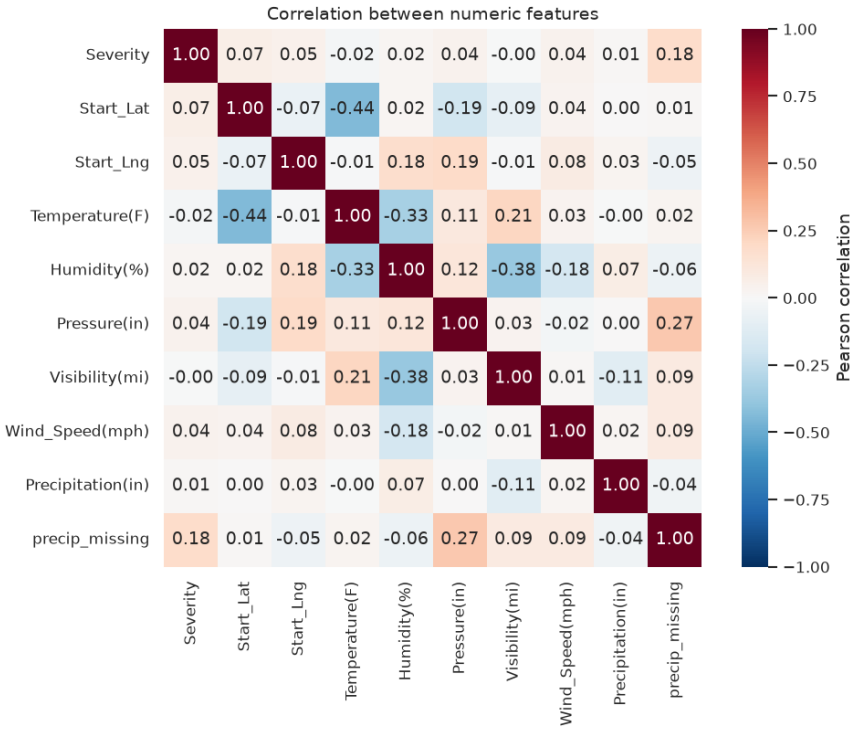


Figure 11. Correlation between numeric features. Pearson correlation; red = positive, blue = negative.

Interpretation: no numeric variable is linearly related to Severity (all |r| below 0.1). The only notable pairs are temperature with latitude (-0.44: the north is colder) and humidity with temperature (-0.33) and visibility (-0.38: fog and rain). No pair is strong enough to make a column useless. The signal is therefore non-linear and comes from combinations of variables, which favours tree-based models later.

4.4 Stability over time and providers (RQ4)

Figure 12 shows the severity mix per year and Figure 13 the role of the three data providers.

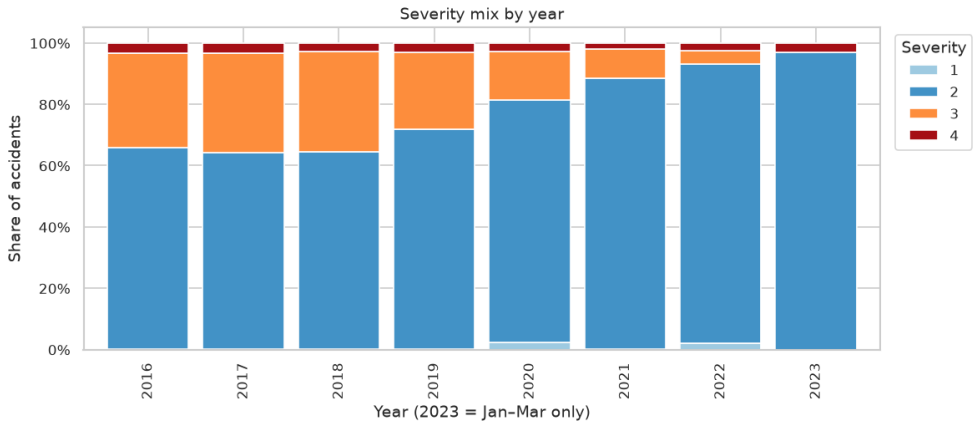


Figure 12. Severity mix by year. Each bar sums to 100%; 2023 = January-March only.

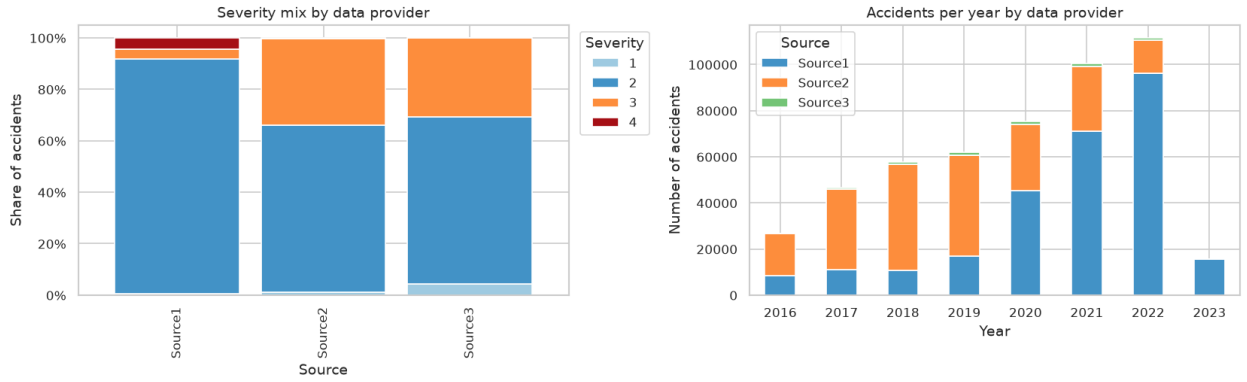


Figure 13. Severity by data provider (left) and accidents per year by provider (right). Left: each bar sums to 100%. Right: stacked counts per year.

Interpretation (RQ4): the label is **not** stable. Severity 3 falls from about 32% of accidents (2017-2018) to 4.5% (2022) and 0% (2023), and Severity 1 appears almost only in 2020 and 2022. Figure 13 explains why: Source2 labels 34% of its accidents as Severity 3, Source1 only 4% (but 4.3% as Severity 4, against 0.4%), and over the years Source1 replaced Source2. Part of "severity" therefore reflects *who recorded the accident*, not only the accident itself. This matters for modelling: a model may learn the provider instead of the accident, and scores on recent years will be lower.

4.5 Hypothesis tests

Charts can show patterns that happen by chance, so we tested them (significance level $\alpha = 0.05$). For two categorical variables we used the **chi-square test of independence** (H0: the variable and Severity are independent; assumption: every expected count is at least 5, which is why the 203 "Other" weather rows were left out). For temperature across the 4 levels we used the **Kruskal-Wallis test** (H0: same distribution in every group) instead of ANOVA, because temperature is not normally distributed (D'Agostino test, $p \approx 10^{-41}$) and the group variances differ (Levene test, $p \approx 10^{-194}$).

With almost 500,000 rows, even tiny differences give $p \approx 0$. So we also report the **effect size** (Cramér's V: 0.1 small, 0.3 medium, 0.5 large; ε^2: 0.01 small, 0.06 medium) and repeat each test on a random sample of 5,000 rows.

Table 11. Hypothesis test results. "≈ 0" means smaller than 10^{-300}. Effect sizes say how strong the relationship is.

H0: Severity is independent of...	Test	Statistic	p-value	p-value (5,000 rows)	Effect size	Decision
Data provider (Source)	Chi-square	$\chi^2 = 82{,}258$	≈ 0	4×10^{-182}	V = 0.29 (small, near medium)	Reject H0
Traffic signal nearby	Chi-square	$\chi^2 = 7{,}336$	≈ 0	8×10^{-13}	V = 0.12 (small)	Reject H0
Day / night	Chi-square	$\chi^2 = 1{,}242$	7×10^{-269}	0.0004	V = 0.05 (negligible)	Reject H0
Junction nearby	Chi-square	$\chi^2 = 1{,}466$	≈ 0	0.13	V = 0.05 (negligible)	Reject H0 (weak)
Weather group	Chi-square	$\chi^2 = 2{,}264$	≈ 0	0.013	V = 0.04 (negligible)	Reject H0
Temperature (same for all levels)	Kruskal-Wallis	H = 1,818	≈ 0	2×10^{-5}	$\varepsilon^2 = 0.004$ (negligible)	Reject H0

Interpretation: every null hypothesis is rejected on the full data, but the effect sizes show what really matters. The **data provider** has by far the strongest relationship with Severity (confirming RQ4), followed by the **road context** (traffic signals). Light conditions, weather and junctions have negligible effects: they are real but weak, and the junction effect is not even significant on 5,000 rows. Temperature differs only slightly between levels. Conclusion: no single variable explains Severity, so a model will have to combine many weak signals.

5. Feature Engineering

Done in notebook `04_features.ipynb` with reusable code in `src/features.py`. There are two kinds of features: features built from a single row (safe to build at any time) and transformations that learn from many rows (fitted on training data only).

5.1 Created features

Table 12. Features created from each row. Last column: share of Severity 3-4 when the feature is 1 vs 0. Highway is the strongest new signal.

New feature	How it is built	Why	High-impact share (yes vs no)
hour, weekday, month, year	From Start_Time	Traffic and reporting change with time	-
season	Month → Winter / Spring / Summer / Autumn	Snow and ice in winter	-
is_weekend	1 if Saturday or Sunday	Different traffic	22% vs 19%
is_rush_hour	1 if weekday and hour in 7-9 or 16-19	Congestion	20% vs 20%
civil_night, astro_night	1 if dark (civil / astronomical twilight)	Darkness, fewer responders	19% vs 20%
weather_group, bad_weather	9 weather groups; 1 if rain, storm, snow/ice or fog	Fewer, stronger categories	21% vs 19%
road_feature_count	Number of road features nearby (0-12)	City street vs open road	22% (0) → 16% (1) → 8% (2+)
is_highway	Street name matches I-95, US-101, CA-99, Hwy, Fwy, Expy, Tpke, Beltway, Route...	Highway accidents block more traffic	28% vs 11%
precip_missing	1 if precipitation was empty (Section 3.3)	Keeps the information that it was guessed	-

The highway rule was checked on the most common street names: the first version missed state routes like "CA-99" and "Capital Beltway"; after fixing it, the unmatched streets are local ones such as "Main St".

5.2 Training split before any fitting

Before selecting or encoding anything, the data is split into **80% training (396,425 rows)** and **20% test (99,107 rows)**, keeping the same Severity shares in both (stratified, seed 42). Every decision below is made on the training part only, so the test part stays unseen, as it will be for a real model.

5.3 Feature selection (RQ5)

We started from 41 candidate inputs and applied three checks on the training data:

- **Near-constant columns:** a yes/no column that is 1 in fewer than 0.1% of rows carries almost no information → Roundabout (0.002%), Bump (0.04%) and Traffic_Calming (0.09%) removed. They are still counted in road_feature_count, so no information is fully lost.
- **Redundant columns:** when two columns are almost the same (|correlation| above 0.85), we keep the one with more information about Severity → is_night removed (r = 0.89 with civil_night) and nautical_night removed (r = 0.88 with astro_night).
- **Mutual information (MI):** measures how much a feature tells us about Severity, including non-linear relations and categories. Used to rank the features (Figure 14) and to choose inside redundant pairs.

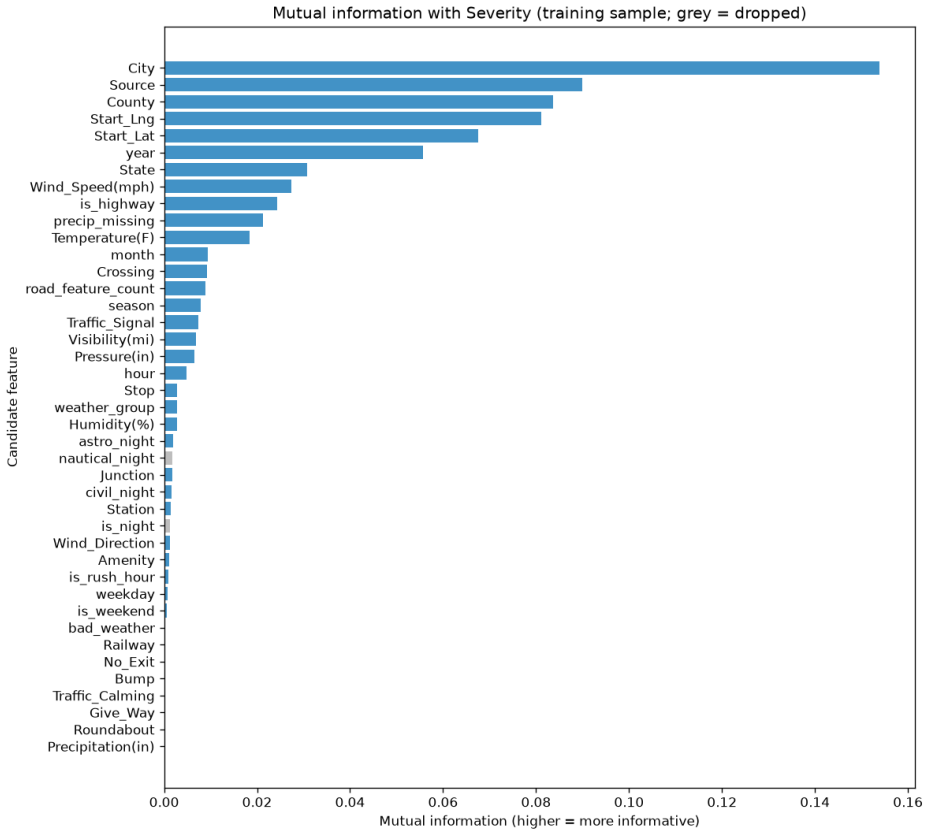


Figure 14. Mutual information between each candidate feature and Severity. Computed on 60,000 training rows; grey bars are the removed features.

Interpretation (RQ5): location (City, County, coordinates, State), the data provider (Source), the year, wind speed and is_highway are the most informative. City and County are partly over-rated because MI is biased upward for columns with thousands of values. Weak features such as weather are kept: the EDA showed they matter a little, and tree models can combine them.

5.4 Encoding and scaling

Table 13. Transformations fitted on the training split only. All are combined in one scikit-learn ColumnTransformer (src/features.py).

Transformation	Columns	What it does	Why this choice
Median filling	Numeric columns	Fills empty values with the training median	Robust to the skewed weather values
One-hot encoding	State, Source, weather_group, Wind_Direction, season	One 0/1 column per value (84 columns)	Few values (at most 49); no false order between them
Frequency encoding	City, County	Replaces each name by its share of training rows; unseen → 0	One-hot would create about 10,000 mostly-empty columns
Location clusters	Start_Lat, Start_Lng	K-means splits the map into 50 regions, then one-hot	Captures regional hotspots that raw coordinates show poorly (Figure 15)
Standard scaling	Numeric columns	Mean 0, standard deviation 1	Only for linear / distance-based models; trees do not need it

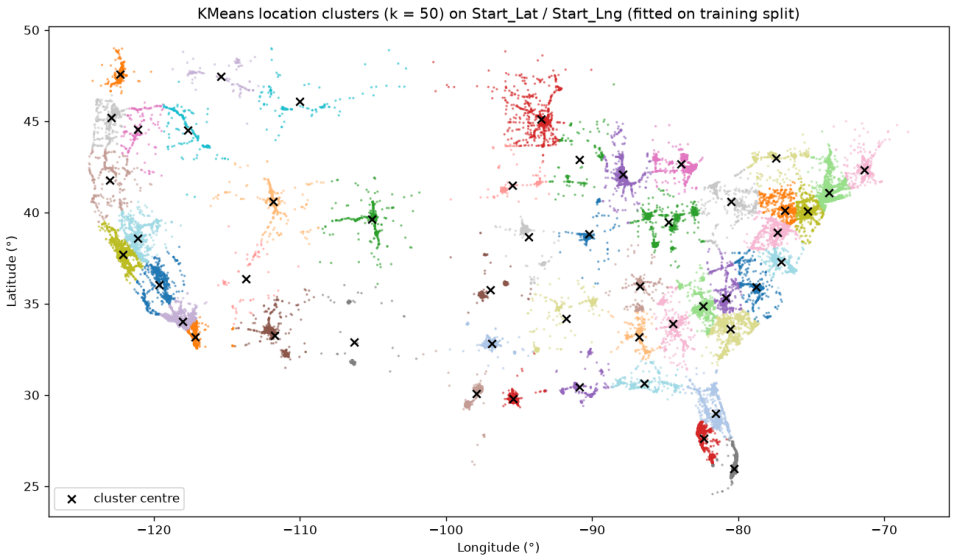


Figure 15. The 50 location clusters. K-means on latitude and longitude of the training accidents; crosses are cluster centres.

5.5 Final feature set and checks

Table 14. Final feature set. Saved in data/features.parquet (495,532 rows × 36 inputs + Severity).

Group	Final inputs (36)	Why kept
Location	Start_Lat, Start_Lng, State, City, County	Most informative; severity varies by region
Road	is_highway, road_feature_count, Junction, Traffic_Signal, Crossing, Stop, Station, Railway, Give_Way, No_Exit, Amenity	Road context has the largest real effect (RQ3)
Time	hour, weekday, month, year, season, is_weekend, is_rush_hour	Time patterns (RQ1); year captures reporting changes
Light	civil_night, astro_night	Night raises Severity 4 (RQ1)
Weather	Temperature, Humidity, Pressure, Visibility, Wind_Speed, Wind_Direction, Precipitation, precip_missing, weather_group, bad_weather	Small effects that may add up (RQ2)
Provider	Source	Strongest link (RQ4); to be tested with and without it

How the features support the objective. Every input is known when the accident is reported, so the model can be used at that moment, which is the goal of Section 1. The created features turn raw columns into the patterns found in the EDA (night, road type, bad weather, rush hour), the selection removes columns that add noise but no information, and the encodings turn text into numbers without creating thousands of empty columns or leaking test data. Source is kept on purpose so that its effect can be measured in the modelling phase (RQ4).

Verification: after fitting on the training split and applying to the test split, both have **165 numeric columns** (29 numeric, 84 one-hot, 2 frequency, 50 regions) and no missing values. The 399 test accidents (0.4%) from cities never seen in training get frequency 0, which proves that nothing was learned from the test data. After scaling, training columns have mean 0 and standard deviation 1, and test columns are close (within 0.01) as expected.

6. Conclusions and Next Steps

6.1 Answers to the research questions

Table 15. Answers to the research questions.

Question	Answer (with evidence)
RQ1 Time	Rush hour brings more accidents but not worse ones. Severe (level 4) accidents are most common at night: 5-6% around midnight vs 1.6% at 7-8 h; 3.5% at night vs 2.2% by day (small effect, V = 0.05).
RQ2 Weather	Small effect: rain and storms raise high-impact accidents from 17% to about 22%; snow/ice raises Severity 4. Statistically significant but negligible in size (V = 0.04); temperature differs only slightly.
RQ3 Road	Clear effect: highway accidents are 28% high-impact vs 11% elsewhere; near traffic signals, crossings and stops only 6-10% (V = 0.12 for signals).
RQ4 Consistency	No: providers label severity very differently (V = 0.29, the strongest link) and the provider mix changes over the years, so severity shifts over time.
RQ5 Features	Location, provider, year and highway are the most informative; 36 inputs kept after removing 3 near-constant and 2 redundant columns.

6.2 Main findings

- The target is very unbalanced (80% Severity 2), so future models need macro F1 and class balancing.
- Leakage was a real risk: 88.5% of Severity 4 descriptions say "closed". Removing these columns makes the task honest, but harder.
- Severity depends more on **where** (road type, region) and **who reported it** than on weather or time.
- No single variable is strongly related to Severity; the signal is weak and non-linear, so tree-based models should do better than linear ones.

6.3 Limitations

- Severity measures traffic delay as recorded by each provider, not injuries, and providers use the scale differently. A model may partly learn the provider's habits instead of the accident's impact.
- We used a 6.5% sample; the data ends in March 2023 and coverage varies by state and year.
- Weather comes from the nearest airport station and may not match the exact place of the accident.
- Some choices are rules of thumb: the impossible-value limits, the highway name rule, the 0.1% and 0.85 selection thresholds, and filling missing precipitation with 0.
- The hypothesis tests show association, not cause.

6.4 Next steps (modelling, not done in this midterm)

- **Splits:** split the training part again into train / validation (70/15/15 overall) for all choices; keep the test set for one final evaluation; add a time-based check (train 2016-2021, test 2022-2023).
- **Imbalance:** compare class weights with undersampling of Severity 2 on validation data.
- **Models:** a trivial baseline (always Severity 2, macro F1 about 0.22) → Logistic Regression → Decision Tree → Random Forest → LightGBM / XGBoost, using the preprocessor of Section 5.
- **Evaluation:** macro F1, recall of Severity 4, balanced accuracy and a confusion matrix.
- **Provider test:** train the best model with and without Source and report the difference.
- **Explanation:** feature importance and SHAP values to check which features drive predictions.

References

[1] Moosavi, S., Samavatian, M. H., Parthasarathy, S., & Ramnath, R. (2019). *A Countrywide Traffic Accident Dataset.* arXiv:1906.05409.

[2] Moosavi, S., Samavatian, M. H., Parthasarathy, S., Teodorescu, R., & Ramnath, R. (2019). *Accident Risk Prediction based on Heterogeneous Sparse Data: New Dataset and Insights.* Proceedings of the 27th ACM

SIGSPATIAL International Conference on Advances in Geographic Information Systems.

[3] Moosavi, S. *US Accidents (2016-2023)* [Dataset]. Kaggle. https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents

[4] Moosavi, S. *US-Accidents: A Countrywide Traffic Accident Dataset.* https://smoosavi.org/datasets/us_accidents

[5] Software: pandas, NumPy, SciPy, scikit-learn, matplotlib and seaborn (Python 3.12).

All figures and tables were produced by the team from the dataset [1-3]; the dataset is used under its CC BY-NC-SA 4.0 licence.

Appendix A: Variable definitions

Table 16. Definition of every column of the raw sample. Type as loaded by pandas; Missing = share of empty cells in the raw sample.

Column	Type	Unit	Definition	Missing
ID	text	-	Unique identifier of the accident record	0.0%
Source	text	-	Data provider that reported the accident (Source1, Source2, Source3)	0.0%
Severity	integer	1-4	Impact on traffic: 1 = short delay, 4 = long delay (target)	0.0%
Start_Time	text	local time	When the accident started	0.0%
End_Time	text	local time	When its impact on traffic ended (known only afterwards)	0.0%
Start_Lat	decimal	degrees	Latitude of the start point	0.0%
Start_Lng	decimal	degrees	Longitude of the start point	0.0%
End_Lat	decimal	degrees	Latitude of the end of the affected road stretch	44.1%
End_Lng	decimal	degrees	Longitude of the end of the affected road stretch	44.1%
Distance(mi)	decimal	miles	Length of road affected by the accident	0.0%
Description	text	text	Free-text description written by the provider	0.0%
Street	text	-	Street name	0.1%
City	text	-	City	0.0%
County	text	-	County	0.0%
State	text	-	US state (2 letters)	0.0%
Zipcode	text	-	Postal code	0.0%
Country	text	-	Country (always US)	0.0%
Timezone	text	-	Time zone of the location	0.1%
Airport_Code	text	-	Airport weather station closest to the accident	0.3%
Weather_Timestamp	text	local time	Time of the weather observation	1.5%
Temperature(F)	decimal	°F	Air temperature	2.1%
Wind_Chill(F)	decimal	°F	Felt temperature with wind	25.8%
Humidity(%)	decimal	%	Relative humidity	2.2%
Pressure(in)	decimal	inHg	Air pressure	1.8%
Visibility(mi)	decimal	miles	How far one can see	2.3%
Wind_Direction	text	-	Wind direction (compass point, CALM or VAR)	2.2%
Wind_Speed(mph)	decimal	mph	Wind speed	7.4%
Precipitation(in)	decimal	inches	Amount of rain or snow	28.5%
Weather_Condition	text	-	Weather description (rain, snow, fog, ...)	2.2%
Amenity	True/False	True/False	Amenity (shop, school, ...) near the accident	0.0%
Bump	True/False	True/False	Speed bump nearby	0.0%
Crossing	True/False	True/False	Pedestrian crossing nearby	0.0%
Give_Way	True/False	True/False	Yield sign nearby	0.0%
Junction	True/False	True/False	Junction or highway exit nearby	0.0%
No_Exit	True/False	True/False	Dead-end road nearby	0.0%
Railway	True/False	True/False	Railway nearby	0.0%
Roundabout	True/False	True/False	Roundabout nearby	0.0%
Station	True/False	True/False	Train / bus station nearby	0.0%
Stop	True/False	True/False	Stop sign nearby	0.0%
Traffic_Calming	True/False	True/False	Speed-reducing road design nearby	0.0%
Traffic_Signal	True/False	True/False	Traffic lights nearby	0.0%
Turning_Loop	True/False	True/False	Turning loop nearby	0.0%
Sunrise_Sunset	text	Day/Night	Day or night, based on sunrise and sunset	0.3%

Column	Type	Unit	Definition	Missing
Civil_Twilight	text	Day/Night	Day or night, based on civil twilight (sun 6° below horizon)	0.3%
Nautical_Twilight	text	Day/Night	Day or night, based on nautical twilight (12°)	0.3%
Astronomical_Twilight	text	Day/Night	Day or night, based on astronomical twilight (18°)	0.3%

Appendix B: How to reproduce

Install the packages in `requirements.txt` and run `python run_all.py` from the project folder. It runs notebooks 01 → 04 in order (about one minute) and recreates every file used in this report: data/sample.parquet, data/clean.parquet, data/features.parquet, reports/leakage_audit.md, reports/cleaning_log.md, reports/data_dictionary.md and reports/figures/. See README.md for setup details.